

UV TEXTURE MAPPING

Think of UV mapping like taking a box and cutting it to lay flat. The texture needs to match each side of the box. Achieving that result with traditional materials and textures is difficult. A digital model would need texture variation to make it look appealing and needs to be mapped correctly. This is where UV mapping comes in.

Create a cube and change the viewport shading mode from solid to textured. This is the shading used during gameplay. Textures are affected by lighting, so add some lights to illuminate your scene better and switch to UV editing window layout. This will provide you with one 3D viewport and one UV editor viewport. At the bottom of the UV, editor viewport select the Image-Open image menu option and select the texture you would like to use.

Now, enter edit mode for the cube and switch to selecting vertices to edges. Firstly, We should mark where the seams would be in our model so that it can be unwrapped properly. Select the edges you wish to cut and select “Mark seam” option from the shading/UV’s tab in the tool shelf. Now, go to face selection mode and hit “A” twice to select all faces. Type “U” to bring up the UV mapping options in the 3D viewport. Select “Unwrap”. You will now see the unwrapped faces in the UV mapping viewport. You will also see the texture applied correctly on the cube.

DISPLACEMENT MAPPING

As mentioned before, Displacements maps affect the silhouette of the mesh and deform the mesh. You can make a cube look wrinkled or deformed using a displacement map. Make sure to subdivide your model a few times so that it has enough divisions to be deformed correctly. Create a cube and add a material to it and add a greyscale texture such as the cloud texture which is inbuilt in blender. Go to the “Influence” panel and find “Displacement” button to turn on displacement and adjust the slider. Render the scene to view the effect of displacement.

REALISTIC MATERIALS

So far, we have applied materials and textures from the materials panel. But the cycles render engine is powerful enough to produce even more realistic results. This power lies within the Node Editor window in blender. Understanding how to connect various nodes and which ones to use at the right place can be a steep learning curve to beginners, and only a few have